

Qiushi Engine

Autonomous Discovery of New Lower Bounds for Kissing Numbers

Codes, Lattice Liftings and Section Counts

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Abstract

During the 2026 Apsara Conference, Zhejiang University and Alibaba Cloud presented a 72-hour livestream of autonomous research by Qiushi Engine on the high-dimensional kissing-number problem. The problem asks how many equal, nonoverlapping spheres can simultaneously touch a central sphere. Constructing larger configurations requires new contact directions that satisfy every pairwise distance constraint. This collection presents improved lower bounds in 10 dimensions obtained through Qiushi Engine's autonomous research. The results cover dimensions 32–39, 43 and 45.

Three related construction methods emerge. In layered codes, joint support exchanges allow compatible candidates to share deletion costs; paired Hadamard transfer builds larger local signed codes and extends their compatibility to the full spherical configuration. The local code in dimension 35 attains the capacity bound of its model, while dimension 37 combines support enlargement with signed replacement. For a Leech-lattice lifting, another shell supplies equatorial directions compatible with every lifted point, giving $K(38) \geq 591900$.

For lattice sections, spherical-design moments turn counts in a large mother shell into finite rational linear problems. An embedded E_8 parent determines the size of a D_5 perpendicular shell, giving $K(43) \geq 2553792$. A moment inequality for the target fibres, combined with common-neighbour search, gives $K(45) \geq 7380090$. These methods connect geometric freedom, discrete selection and exact proof. Each of the stated bounds has an explicit mathematical construction and reconstructible finite data.

Keywords Kissing number; constant-weight code; Hadamard transform; Leech lattice; lattice section; spherical design

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1 The problem and main results

The kissing-number problem asks for the largest number $K(n)$ of unit balls that can touch a central unit ball in Euclidean n -space without overlapping interiors. Dividing the outer centres by two gives an equivalent problem on the unit sphere: place as many points as possible at angular distance at least 60° :

$$K(n) = \max\{|X| : X \subset S^{n-1}, \langle x, y \rangle \leq \frac{1}{2} \quad (x \neq y)\}.$$

The equivalence follows from $|x - y|^2 = 2 - 2\langle x, y \rangle$. It connects local sphere packing to spherical codes, and makes lattices, error-correcting codes and finite geometry natural sources of constructions [1–3].

Higher dimension allows more directions, but the number of pair constraints grows quadratically with the size of a configuration. An additional point may obstruct many old points, so searching for an isolated gap often fails to improve a large construction. Lattices and codes organize these constraints: sign patterns on a support share one distance argument, and vectors in a shell share norm and product restrictions. The problem becomes how to exploit that organization without freezing every choice it contains. We retain structures controlling most pairs while changing finite parts that determine the cardinality.

1.1 Results

Theorem 1.1. *For every dimension n in Table 1, there is a kissing configuration of N_n points, where N_n is the lower bound in the second column. Thus $K(n) \geq N_n$.*

The constructions in this report give the following lower bounds. The comparison column uses the fixed public sources recorded on 24–25 September 2026.

Table 1 Explicit lower bounds and their public comparison values.

Dimension	Lower bound	Public comparison	Increase	Source
32	347584	346432	1152	[4]
33	363968	362048	1920	[4]
34	384196	381124	3072	[4]
35	409676	409548	128	[4]
36	484760	484568	192	[4]
37	498024	496232	1792	[4]
38	591900	591612	288	[5]
39	763668	756116	7552	[5]
43	2553792	2545056	8736	[6, 7]
45	7380090	7379838	252	[6, 7]

1.2 Background on the constructions

The first route starts with the three-layer construction of Edel, Rains and Sloane [8]. Its count combines a dense binary code, a family of weight-eight supports and a root system. The Cheng–Sloane code supplies the length-32 dense layer [9]. Constant-weight constructions and comparisons are developed in the work of Brouwer, Shearer, Sloane and Smith and in the subsequent tables [4, 10]. Echols recently improved the bounds in dimensions 32, 33, 34 and 37 by support

search [11]. Two further choices remain: can candidates share the cost of deleting old supports, and must a complete sign bundle remain indivisible? The local replacement principle has a direct precursor in the free zones of Takhanov–Yun [12, Section V]. The constructions here combine that principle with paired Hadamard transfer and explicit ERS embeddings. The seven code-based results give the support families, transverse selections and boundary conditions needed to turn these choices into larger configurations.

The second route uses the minimal shell of the Leech lattice. A lifting combines a direction in the mother space with a low-dimensional tail; the two products together determine compatibility. The lifting framework of Cohn, Jiao, Kumar and Torquato [13, Theorem 7.5] and subsequent constructions show how directions can be reused. Our dimension-38 result starts from Kravatskiy’s public 591612-point configuration [5]. All minimal Leech lines already occur in its lifted layer, but its equator is empty. Instead of reusing those same minimal directions again, the search moves to another shell and seeks directions compatible with every lifted point. This adds 288 points.

The third route takes a small section of an extremal 48-dimensional lattice and studies its perpendicular shell or the projections of selected fibres. Conway–Sloane sections and the antipode construction establish this connection [14, 15]; the exact counts of Sun and Wang supply the public comparisons in dimensions 43 and 45 [6, 7]. The mother shell has 52416000 vectors. The later analysis of embedded E_8 sections by Dorofeev, Sun and Wang [16] provides the relevant parent geometry for the 43-dimensional calculation. The large shell makes direct enumeration expensive. Moreover, a section’s abstract root type and determinant do not determine its embedding. Spherical-design theory [2, 17] supplies exact low-degree moment identities constraining projection multiplicities. We use them to determine a 43-dimensional section count and to turn a small explicit witness set into a lower bound of millions of points in dimension 45.

1.3 Main contributions and construction principles

The research gives explicit lower bounds in 10 dimensions and mathematical methods linking construction searches to exact proofs. In layered codes, shared conflicts determine the net gain of a joint support exchange. Paired Hadamard transfer turns weight-four codes into compatible local weight-eight signed codes, allowing support selection and sign assignments to change together. The 256-point replacement in dimension 35 attains the capacity bound of its eleven-coordinate model; the 576-point replacement in dimension 36 has a direct four-colour construction. Dimension 37 combines support enlargement and signed replacement, with gains of 1664 and 128 points. Each construction includes an explicit finite object and a proof of its boundary compatibility.

In dimension 38, the key is to identify geometric space still available outside the lifted layer. We select 144 compatible lines from another Leech shell and prove both their mutual angular separation and their compatibility with every lifted point. This preserves the base configuration and adds 288 equatorial points. The construction resolves the choice of directions, the inner-product constraints between shells and the common coordinate realization of both layers.

Dimensions 43 and 45 turn geometric construction problems into exact counting problems. In dimension 43, an embedded E_8 parent constrains the fibre multiplicities of a D_5 child; rational moment identities determine a count of 2,553,792 points. In dimension 45, a rational inequality for the target fibres is combined with a common-neighbour graph for witness search. The resulting 298 explicit vectors establish a lower bound of 7,380,090. These constructions exhibit

two distinct uses of counting: determining a child's exact size from its parent structure, and using a small constructible set to control the size of a large projected configuration.

Qiushi Engine autonomously conducted the construction searches, mathematical analysis and computational verification. Its autonomous experimental research is described in [18]. After the result table, the report proceeds through code constructions, equatorial completion and section counts. Each chapter introduces the object being changed, derives its compatibility conditions, and explains how the finite data specify the resulting construction.

2 Layered codes, joint exchanges and signed constructions

These seven results use one geometric construction, with improvements at different levels. Dimensions 32, 33 and 34 use larger compatible families of eight-element supports; dimension 39 combines such a family with an explicit sign code. Dimensions 35, 36 and 37 also change the signs carried by the supports. The important transition is from deciding whether to retain a whole bundle to replacing it by a different finite set.

2.1 Compatibility of the layers

The Edel–Rains–Sloane construction [8] places three coordinate patterns on the same unit sphere. Let $n \geq m \geq 32$, let $T \subset \{0, 1\}^m$ be a binary code, and let $\mathcal{B}$ be a family of eight-element subsets of $\{0, \dots, n-1\}$. A support records the nonzero coordinates of a vector; the signs specify the vector itself. The dense, middle and root layers are

$$\begin{aligned} X_T &= \left\{ \frac{1}{\sqrt{m}}((-1)^{c_0}, \dots, (-1)^{c_{m-1}}, 0, \dots, 0) : c \in T \right\}, \\ X_{\mathcal{B}} &= \left\{ \frac{1}{\sqrt{8}} \sum_{i \in B} \varepsilon_i e_i : B \in \mathcal{B}, \varepsilon_i \in \{-1, 1\}, \prod_{i \in B} \varepsilon_i = 1 \right\}, \\ X_R &= \left\{ \frac{\pm e_i \pm e_j}{\sqrt{2}} : 0 \leq i < j < n \right\}. \end{aligned}$$

Each support carries $2^7 = 128$ even-parity sign patterns. The layers have respectively m , eight and two nonzero coordinates, so they are disjoint. Their union contains

$$N = |T| + 128|\mathcal{B}| + 2n(n-1) \quad (2.1)$$

points.

Proposition 2.1 (Three-layer construction [8]). *If T has minimum Hamming distance at least $\lceil m/4 \rceil$ and distinct supports satisfy $|B \cap B'| \leq 4$, then the union is a kissing configuration in dimension n .*

Proof. Two dense-layer points have product $1 - 2d_H(c, c')/m \leq 1/2$. Distinct even-parity patterns on the same support differ in at least two positions, giving at most $1 - 4/8 = 1/2$. Different supports give at most $|B \cap B'|/8 \leq 1/2$. Distinct roots either share at most one coordinate with the same sign, or have the same support and differ in at least one sign; their product is also at most $1/2$.

There are three remaining cross-layer cases. Dense–middle products are at most $8/(\sqrt{m}\sqrt{8}) = \sqrt{8/m} \leq 1/2$; dense–root products are at most $2/(\sqrt{m}\sqrt{2}) = \sqrt{2/m} \leq 1/2$; middle–root products are at most $2/(\sqrt{8}\sqrt{2}) = 1/2$. These cover all distinct pairs. $\square$

The geometric problem reduces to finding a large binary code and a large support family with intersection at most four. The support indicators have constant weight eight and satisfy $d_H(1_B, 1_{B'}) = 16 - 2|B \cap B'|$. Thus the intersection condition is minimum distance eight in the constant-weight code. Equation (2.1) converts improvements of such codes into kissing-number lower bounds [4, 10, 11].

2.2 Joint exchanges of supports

Inserting a new support generally requires removing old supports with large intersection. For a single candidate, these deletions may consume the gain. Several mutually compatible candidates can instead be obstructed by the same old supports. The search therefore selects additions jointly.

For a candidate b , put $F(b) = \{B \in \mathcal{B} : |B \cap b| > 4\}$. If Y is disjoint from the old family and its internal intersections are at most four, then

$$\begin{aligned} \mathcal{B}' &= (\mathcal{B} \setminus F(Y)) \cup Y, & F(Y) &= \bigcup_{b \in Y} F(b), \\ |\mathcal{B}'| - |\mathcal{B}| &= |Y| - |F(Y)|. \end{aligned} \tag{2.2}$$

The cost is a union size, not the sum of the individual conflict counts. The stored final exchange in dimension 33 removes nine supports and adds ten; in dimension 34 it removes six and adds seven; an exchange in dimension 37 removes 37 and adds 40. The data contain the parent families and actual deletions and additions, so these unions can be reconstructed.

There is also a direct discrete formulation. For a finite candidate set $\mathcal{C}$, let $z_b \in \{0, 1\}$ select a candidate and let $u_B \in \{0, 1\}$ delete an old support. Impose $z_b + z_{b'} \leq 1$ for incompatible candidates and $z_b \leq u_B$ for $B \in F(b)$, and maximize

$$\sum_{b \in \mathcal{C}} z_b - \sum_{B \in \mathcal{B}} u_B.$$

An optimal choice deletes only supports actually blocking selected candidates, so its objective is exactly the gain in (2.2). This identifies the useful search property: concentrate conflicts on shared old supports, rather than minimizing each candidate's conflict count separately. Candidate generation and joint selection are complementary tasks; changing the candidate family changes the constructions available to this finite optimization problem.

The blocking sets also organize candidate generation. For a chosen integer f , consider eight-subsets outside $\mathcal{B}$ with $|F(b)| \leq f$. If the pool includes all such subsets, it covers every positive exchange with at most $f + 1$ additions: for $b \in Y$, $|F(b)| \leq |F(Y)| < |Y| \leq f + 1$. The threshold therefore describes a specific exchange neighbourhood rather than a limit on the final code.

Candidate conflicts admit a compact representation. Two eight-subsets intersect in at least five positions exactly when they share a five-subset. Each candidate occupies $\binom{8}{5} = 56$ such subsets, and the constraints are

$$\sum_{b \in \mathcal{C}: Q \subset b} z_b \leq 1 \quad \left(Q \in \binom{\{0, \dots, n-1\}}{5} \right).$$

They may be added successively: solve the current selection problem, inspect the selected five-subsets, and add the constraints at repeated subsets. Once none repeats, the selection is compatible and a positive objective gives an exchange. The recorded dimension-34 pool contains

10715 candidates with at most three blockers; this procedure yields the seven-for-six exchange above. After each accepted exchange, recomputing the blocking sets makes further replacements available.

The following parameters give direct three-layer constructions. The dimension-37 support family was enlarged from 2843 to 2845 and then modified by a signed replacement, whose final count is given below.

Table 2 Parameters of the direct three-layer constructions.

n	Dense length m	$ T $	$ \mathcal{B} $	Points
32	32	131072	1676	347584
33	32	131072	1803	363968
34	32	131072	1960	384196
39	39	327680	3383	763668

Each extra support contributes 128 spherical points. For example, $131072 + 128 \cdot 1676 + 1984 = 347584$ in dimension 32. The gains of 15 and 24 supports in dimensions 33 and 34 give respectively 1920 and 3072 points over the comparison constructions.

The length-32 layer uses the Cheng–Sloane [32, 17, 8] code [9], whose generator expands to $2^{17} = 131072$ words. Dimension 39 uses $T = \bigcup_{a=1}^{10} (H + r_a)$, where H has dimension 15 and minimum weight 12. Enumeration of the supplied kernel and representatives gives

$$\min_{h \in H} \text{wt}(h + r_a + r_b) = 10$$

for all 45 distinct coset pairs. The cosets are disjoint, with $10 \cdot 2^{15} = 327680$ words and minimum distance ten, as required. The kernel is obtained by shortening an extended quadratic-residue code; the programs compare generator row spaces and check the actual distances.

The listed dimension-39 comparison has the same dense-layer size. Increasing its 3324 supports to 3383 adds $128(3383 - 3324) = 7552$ points. Recording the support and dense layers separately identifies the finite objects and their respective contributions.

2.3 Hadamard transfer and local replacement

The support count is not the only variable. Each original support carries all 128 even-parity patterns. Removing a few complete bundles and inserting a different signed set can fill the affected region more efficiently. Both the internal products and products against the retained configuration must be controlled.

Proposition 2.2 (Paired Hadamard transfer). *Let $\mathcal{A}$ be a family of four-element supports with intersections at most two, transverse to a fixed coordinate pairing. Applying a Hadamard matrix to each pair in all signed versions of $\mathcal{A}$ gives $16|\mathcal{A}|$ weight-eight vectors of squared norm eight and distinct-pair products at most four.*

Proof. Start with four-element supports having pairwise intersections at most two, and attach all sixteen sign patterns to each. Pair the coordinates and retain supports meeting each pair in at most one coordinate; call them transverse. Apply

$$H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

to each coordinate pair. Both $(\varepsilon, 0)$ and $(0, \varepsilon)$ become two nonzero coordinates, so weight four becomes weight eight. Distinct input vectors have product at most two: different supports satisfy the intersection condition, and distinct sign patterns on the same support differ in at least one position. Since $H^T H = 2I$, the images have squared norm eight and pairwise products at most four. Invertibility preserves their number. $\square$

After division by $\sqrt{8}$, these vectors form an internally compatible spherical set Q . Intersections at most four with each retained support ensure compatibility with the retained middle layer. The dense and root cross bounds remain $\sqrt{8/32} = 1/2$ and $1/2$, independently of the parity of the new signs. This is why replacements need not consist of complete even-parity bundles.

The free-zone replacement principle appears in Takhanov–Yun [12, Section V, Definition 5 and Lemma 4] for signed weight-four codes. Here it is applied to weight-eight regions inside the ERS middle layer. The choices of region, transverse supports and paired transform give the replacements below. In dimensions 35 and 36, boundary compatibility can be secured before choosing Q . Take a coordinate region W , remove the old supports contained in it, and require $|B \cap W| \leq 4$ for every retained support. Any new weight-eight vector on W is then compatible with the retained middle layer. Only its local internal products remain to be controlled. The regions are $W = \{12, \dots, 22\}$ and $W = \{12, \dots, 23\}$, respectively, with zero-based coordinates. They contain one and three old supports. The deletion costs are fixed at 128 and 384 points, turning a problem involving the whole configuration into a smaller independent construction problem.

Direct search among signed weight-eight vectors in eleven and twelve coordinates has respectively $2^8 \binom{11}{8} = 42240$ and $2^8 \binom{12}{8} = 126720$ candidates. Weight-four transfer replaces this search with the choice of a combinatorial source and a coordinate pairing, while preserving an algebraic proof of the inner-product bounds. Dimension 35 starts with a Steiner quadruple system $S(3, 4, 10)$: each triple belongs to exactly one block, giving $\binom{10}{3}/4 = 30$ blocks with pairwise intersections at most two. Enumerating the 945 perfect matchings on ten coordinates yields sixteen transverse blocks. For dimension 36 we use the 51-block construction of Kalbfleisch–Stanton [19, Theorem 7], also described in [12, Section V, Construction 1]. Split twelve coordinates into two sets of six and use the same five one-factors of K_6 in each half. For every colour, the nine choices of a left and a right edge give a four-element block, producing 45 crossing blocks. Add, in each half, the complements of the three edges of the first colour to obtain six further blocks.

The resulting 51 blocks still have intersections at most two. Same-colour crossing blocks sharing one edge intersect in two points; different colours share at most one endpoint in each half. Internal blocks in the same half intersect in two points; those in different halves are disjoint. An internal block meets a crossing block in at most two points.

Proposition 2.3 (A 576-point local code). *The paired Hadamard image of 36 transverse blocks from the two one-factorizations is a 576-point weight-eight code in twelve coordinates.*

Proof. Choose the three edges of one colour as the coordinate pairs in each half. Each edge of the other four colours meets every chosen pair at most once. Its union with a same-colour edge from the other half is therefore transverse. There are $4 \cdot 3 \cdot 3 = 36$ such blocks; their intersections are at most two by the preceding argument. Proposition 2.2 gives $36 \cdot 16 = 576$ points. The six internal blocks are unnecessary for this transverse subfamily. $\square$

In the supplied local coordinate order, the chosen pairs are

$$(0, 1), (2, 4), (3, 5), (6, 7), (8, 10), (9, 11).$$

They form the same colour in the two halves. The direct construction with these pairs regenerates exactly the stored 576 vectors. The accompanying program also enumerates all 945 and 10395 matchings of the two source codes, recovering the 256-point and 576-point sets. Thus the existence proof is explicit, while the exhaustive calculation records the finite choices available in this model.

The three constructions select 16, 36 and 32 transverse supports, producing 256, 576 and 512 new points. The compatible region in dimension 35 has eleven coordinates; ten form five pairs and the remaining coordinate is zero. Dimensions 36 and 37 use twelve coordinates each. The explicit pairs and transverse supports regenerate all 256, 576 and 512 signed points for comparison with the final sets.

Table 3 Signed replacements. The support count precedes the deletion of bundles.

Dimension	Original supports	Bundles removed	Points added	Net gain
35	2157	1	256	128
36	2742	3	576	192
37	2845	3	512	128

Combining the support choices and replacements gives

$$\begin{aligned} K(35) &\geq 131072 + 128(2157 - 1) + 256 + 2(35)(34) = 409676, \\ K(36) &\geq 131072 + 128(2742 - 3) + 576 + 2(36)(35) = 484760, \\ K(37) &\geq 131072 + 128(2845 - 3) + 512 + 2(37)(36) = 498024. \end{aligned}$$

Relative to the comparison with 2832 supports, dimension 37 gains 1664 points from thirteen extra supports and 128 from the replacement, totalling 1792. Dimensions 35 and 36 keep their original support counts; their gains come entirely from the signed choices.

The local choice in dimension 35 also attains an exact capacity bound. Let $M(11)$ be the largest number of weight-eight vectors in $\{0, \pm 1\}^{11}$ with distinct-pair products at most four. Fix a full sign string $\sigma \in \{\pm 1\}^{11}$. Two vectors agreeing with σ on their supports would intersect in at least five positions and have product at least five. Thus each sign string extends at most one selected vector. Each vector, in turn, has eight extensions on its three zero coordinates. Counting gives

$$8|Q| \leq 2^{11}, \quad M(11) \leq 256.$$

The 256-point construction attains equality, so $M(11) = 256$. This explains the complete use of the eleven-coordinate local region. In twelve coordinates, two eight-element supports can intersect in only four positions, allowing a sign string to extend several vectors; this is the geometric opening used by the larger local code in dimension 36.

The finite checks use the coordinate order, pairs and four-subsets specified in the data. In dimension 37, a support string numbered from one at the left has coordinate i at integer-mask bit $37-i$. The program regenerates all 512 points from 32 transverse supports and six coordinate pairs, checks equality with the stored replacement, and checks intersections with every retained support. The origin, count and geometric conditions are thereby connected through the same input.

These seven results treat the support family and its sign patterns as coupled choices. The next construction changes another part of the geometry: equatorial directions left available by a large lattice lifting.

3 Equatorial completion of a Leech-lattice lifting

The signed replacements balance additions against deletions. Here a large lifting leaves an unused equator. We seek directions that can be inserted directly, compatible both with one another and with the entire lifted configuration. A set of 144 such lines contributes 288 points, increasing the lower bound from 591612 to 591900.

3.1 The lifted configuration and its equatorial constraints

Let $\Sigma \subset \mathbb{Z}^{24}$ be the squared-norm-32 minimal Leech shell, with 196560 vectors. The supplied extended Golay code generates 1104 vectors with two entries ± 4 ; 97152 vectors with entries ± 2 on a weight-eight support and even sign parity; and 98304 vectors obtained from signs $(-1)^{c_i}$ by reversing one chosen sign and multiplying that entry by three. The Golay generator has rank 12 and weight enumerator $1 + 759z^8 + 2576z^{12} + 759z^{16} + z^{24}$. Distinct vectors in this standard shell model have inner product at most 16 [1].

Write $\mathbb{R}^{38} = \mathbb{R}^{24} \oplus \mathbb{R}^{14}$. The public 591,612-point construction uses every minimal Leech direction in a cap layer [5], but leaves the equatorial subspace available for other directions. The new construction exploits this distinction.

Partition the 98,280 antipodal lines of Σ into 644 classes whose representatives satisfy $|u \cdot u'| \leq 8$ within each class. In $\mathbb{R}^{14}$ take 1932 vectors of squared norm 8 and distinct inner products at most 4, partitioned into 644 zero-sum triangles. Each triangle has pairwise products -4 . Pair a head class with a triangle and form six points per head line:

$$\left(\frac{\varepsilon u}{\sqrt{48}}, \frac{d}{\sqrt{24}} \right), \quad \varepsilon \in \{1, -1\},$$

with d ranging over the assigned triangle. These are unit vectors. Equal heads with different tails give product $32/48 - 4/24 = 1/2$; different heads with an equal tail give at most $8/48 + 8/24 = 1/2$. Points from different classes give at most $16/48 + 4/24 = 1/2$, and the other cases give smaller bounds. The cap layer has $6 \cdot 98280 = 589680$ points.

Both signs on a head line are included. Opposite heads contribute $-32/48$, so even with identical tails their total product is only $-1/3$.

Add a pure tail layer $(0, Rd/\sqrt{8})$, using the same 1932 directions and an exact rational rotation $R = Q/D$. Its integer data satisfy

$$QQ^\top = D^2 I, \quad |d \cdot Qd'| \leq \sqrt{48} D.$$

The first identity preserves the pure tail products; the second bounds cap-tail products by $1/2$. Integer square roots suffice to check the latter exactly. This gives the base count $589680 + 1932 = 591612$.

3.2 Compatible lines in the second shell

We use the norm-48 shell of the same Leech lattice. If $\mathcal{G}$ is the supplied Golay code, its integer coordinate model is

$$\mathcal{L} = \left\{ x \in \mathbb{Z}^{24} : \begin{array}{l} x_i \equiv m \pmod{2} \text{ for some } m \in \{0, 1\}, \\ ((x_i - m)/2 \bmod 2)_{i=1}^{24} \in \mathcal{G}, \\ \sum_i x_i \equiv 4m \pmod{8} \end{array} \right\}. \quad (3.1)$$

Linearity of $\mathcal{G}$ and its all-one word make this set a lattice. Its minimum squared norm is 32. Indeed, for even coordinates a nonzero Golay residue has at least eight entries of absolute value two. A zero residue makes all coordinates multiples of four, and the sum congruence excludes a single entry of absolute value four. For odd coordinates, a vector of squared norm below 32 would have every entry ± 1 ; the doubly even Golay weight then makes its coordinate sum zero modulo eight, contrary to the last congruence. The minimal vectors described above attain 32.

Lemma 3.1 (Compatibility of the second shell). *If $v \in \mathcal{L}$ has squared norm 48 and $u \in \mathcal{L}$ has squared norm 32, then $|v \cdot u| \leq 24$.*

Proof. The distinct norms imply $v \pm u \neq 0$. Their squared norms are $80 \pm 2v \cdot u \geq 32$, which gives the claimed bound. $\square$

Although all minimal lines are used, equatorial points can come from directions of another length. Squared norm 48 gives the same normalization denominator as the cap heads. The new compatibility conditions consequently become integer-product checks with threshold 24. The 144 supplied vectors satisfy all three congruences in (3.1). It remains to check their mutual products:

$$\|v_i\|^2 = 48, \quad |v_i \cdot v_j| \leq 24 \quad (i \neq j), \quad |v_i \cdot u| \leq 24 \quad (u \in \Sigma).$$

Proposition 3.2 (Equatorial completion). *Adding the 288 points $(\pm v_i/\sqrt{48}, 0)$ to the cap and pure-tail layers gives a 591900-point kissing configuration in dimension 38.*

Proof. The 288 new equatorial points $(\pm v_i/\sqrt{48}, 0)$ are internally compatible. Their products with a cap are $\pm v_i \cdot u/48 \leq 1/2$, and they are orthogonal to the pure tails. No base point lies on the equator, so no deletion is needed. Thus

$$K(38) \geq 591612 + 2 \cdot 144 = 591900.$$

$\square$

Lemma 3.1 removes the entire old shell from the search constraints. Generate integer norm-48 vectors satisfying the lattice congruences and retain one canonical representative per antipodal line. Join two representatives when $|v \cdot v'| > 24$. The remaining problem is an independent-set problem in this finite graph: each selected line contributes exactly two equatorial points. A joint replacement deletes the union of the old neighbours of the inserted lines, so it can improve the set even when no single line can be added. Every accepted set is checked for mutual compatibility. The verifier also recomputes all products with Σ , independently checking the cross-shell lemma on the supplied data; these products need not be recalculated during subset selection.

There is a direct geometric explanation. Write $C = \Sigma/\sqrt{32}$ for the unit mother shell. An equatorial direction $z = v/\sqrt{48}$ must satisfy $|z \cdot c| \leq \sqrt{3/8}$. This threshold is strictly larger than

1/2, because the head of a mixed point has length $\sqrt{2/3}$. The space available at the equator of the lifting is therefore larger than the space for inserting unit points directly into the mother shell. The second-shell directions exploit this space. Using every minimal direction need not exhaust the possible points in the resulting 38-dimensional configuration.

The search first produced 142 compatible lines and then 144. The completed data use the same Golay basis as the cap construction. Canonical line representatives have positive first nonzero coordinate and are sorted lexicographically. This convention binds the class labels, tail triangles and added-line checks to one coordinate system.

4 Extremal lattice sections and design moments

The preceding chapters specify each layer and check its products. Here we select vectors from a large known lattice shell and obtain lower-dimensional configurations by projection. The required information is the number of vectors with particular projection signatures. Dimension 43 counts shell vectors perpendicular to a five-dimensional section; dimension 45 projects three fibres of a three-dimensional section. Both become finite multiplicity problems.

4.1 Mother shells, projection domains and moments

An extremal even unimodular lattice in dimension 48 has minimum squared norm six. Its theta series is

$$E_4^6 - 1440E_4^3\Delta + 125280\Delta^2 = 1 + 52416000q^3 + O(q^4).$$

Thus its minimal shell has 52416000 vectors. Here $E_4 = 1 + 240q + 2160q^2 + 6720q^3 + \dots$ and $\Delta = q - 24q^2 + 252q^3 + \dots$; the constant term and vanishing first two coefficients determine the expression in weight 24 [1]. Venkov's theorem makes this shell a spherical 11-design [17, Theorem 4.1].

A spherical 11-design has the same averages as the sphere of equal radius for every polynomial of degree at most eleven. Thus its low-degree moments along any chosen direction are known without listing its vectors. For example,

$$\sum_{x \in X} (x \cdot t)^2 = \frac{6|X|}{48} \|t\|^2, \quad \sum_{x \in X} (x \cdot t)^4 = \frac{3 \cdot 6^2 |X|}{48 \cdot 50} \|t\|^4.$$

Grouping vectors by projection turns the left sides into linear combinations of unknown multiplicities, with completely known right sides.

For section vectors t_i with Gram matrix H , a shell vector has signature $y_i = x \cdot t_i$ and projection norm $y^\top H^{-1} y \leq 6$. The signatures are integral when the section vectors lie in the lattice. Inner-product bounds against known short section vectors further restrict this finite domain. For degree $2m \leq 10$,

$$\sum_{x \in X} \prod_{\ell=1}^{2m} (x \cdot t_{i_\ell}) = \frac{|X| 6^m}{48 \cdot 50 \cdots (48 + 2m - 2)} \sum_{\mathcal{P}} \prod_{(a,b) \in \mathcal{P}} H_{i_a i_b},$$

where $\mathcal{P}$ ranges over pairings of the labelled positions; degree zero gives the shell size. Unknown signature multiplicities therefore satisfy rational linear equations. In dimension 43 these

equations determine the target statistic exactly. In dimension 45 they are combined with non-negativity to bound the statistic from below.

4.2 Dimension 43: counting a child section through its parent

Let X be the minimal shell of P_{48n} [20]. A supplied integer matrix embeds $\sqrt{3}E_8$ into this lattice. Its first five simple roots span a $\sqrt{3}D_5$ section U , and $X \cap U^\perp$ lies in dimension 43.

The root numbering is fixed by the following standard coordinates:

$$\alpha_1 = \frac{1}{2}(1, -1, -1, -1, -1, -1, -1, 1), \quad \alpha_2 = e_1 + e_2, \quad \alpha_j = e_{j-1} - e_{j-2} \quad (3 \leq j \leq 8).$$

The supplied 8×48 integer matrix T and ambient Gram matrix G satisfy $(TGT^\top)_{ij} = 3\alpha_i \cdot \alpha_j$; rows 1, 2, 3, 4, 5 specify the child. This fixes the embedding as well as the abstract D_5 root type. The use of an embedded $\sqrt{3}E_8$ parent is developed in Dorofeev–Sun–Wang [16]; here its projection data are used to count this D_5 complement.

Proposition 4.1 (The specified D_5 complement). *For the matrices G, T supplied in the certificate, the shell $X \cap U^\perp$ contains exactly 2553792 vectors. Consequently $K(43) \geq 2553792$.*

No direction change is needed. Distinct minimal vectors satisfy $\|x - y\|^2 \geq 6$, so $x \cdot y \leq 3$; division by $\sqrt{6}$ gives a kissing configuration. The task is to count $|X \cap U^\perp|$ exactly.

For a minimal vector, its projection onto the parent section is $\lambda/\sqrt{3}$ with $\lambda \in E_8$. The norm bound and the inner products with every embedded root give a complete domain of 26,641 possible signatures. Of these, 240 are exceptional signatures of the section roots themselves.

This domain exhausts the possible projections. Nonexceptional λ satisfy $\|\lambda\|^2 \leq 18$ and $|\lambda \cdot r| \leq 3$ for every E_8 root r . Enumeration gives shells of squared norm 0, 2, 4, 6 and 8, of sizes 1, 240, 2160, 6720 and 17280 respectively. Adding the 240 exceptions $\lambda = 3r$ gives the complete domain.

The extremal 48-dimensional shell is an 11-design. For projection coordinates, every even moment through degree ten is therefore the corresponding spherical moment. Averaging over the parent Weyl group yields multiplicities

$$1092000, \quad 116235, \quad 9180, \quad 495, \quad 63/4$$

on its five nonexceptional norm orbits. As orbit averages of integer fibre sizes, these multiplicities can be rational.

The perpendicular child has 1, 12, 6, 24, 0 signatures on those orbits and twelve exceptional roots. Hence the average child count is

$$1092000 + 12(116235) + 6(9180) + 24(495) + 12 = 2553792.$$

Thus at least one child has at least 2,553,792 points. To count the specified child exactly, average only over a subgroup preserving that child. Explicitly, use the group generated by $-I$, the reflections in $\alpha_1, \dots, \alpha_5$, and the reflections in the twelve E_8 roots perpendicular to all five. The latter roots form A_3 ; each reflection is $s_r(x) = x - (x \cdot r)r$. These formulas determine the group without reference to an ambient automorphism. The complete domain has 43 orbits under this group. Rational moment rows form a matrix A , with right side b and target c . The certificate gives w satisfying

$$A^t w = c, \quad w^t b = 2553792.$$

This proves Proposition 4.1. The subgroup need not act by automorphisms of the whole mother lattice: it preserves the signature domain, the moment equations, and the target statistic.

The calculation determines a linear statistic rather than every fibre. Even if $Av = b$ has several solutions, $c^T v$ is constant on them whenever c^T lies in the row space of A . Independent moment rows can first be selected by modular linear algebra, followed by a rational solution for the multiplier. Reproduction then requires just the two displayed rational identities. The complete signature domain remains essential: the moments on the right describe the whole shell, while orthogonality belongs in the target vector c , not in a preliminary deletion of all other fibre columns.

The comparison section has the same abstract D_5 Gram type. What changes is the embedding, which can change the perpendicular population. Its extension to an E_8 parent supplies additional projection constraints that certify this population. The additional equations determine a count of the fixed set; they do not change that set.

4.3 Dimension 45: 298 witnesses control the projected count

Use a specified quadratic-residue even neighbour of rank 48. The code-distance and parity argument in Appendix A proves its minimum squared norm six. Its shell again has 52,416,000 vectors and is an 11-design.

Three section vectors have Gram matrix

$$H = 8I_3 - 2J_3, \quad H^{-1} = (I_3 + J_3)/8.$$

Let $f(y)$ count shell vectors of signature y . The nonexceptional domain is explicitly

$$D_0 = \{y \in \mathbb{Z}^3 : |y_i| \leq 3, \quad |y_1 + y_2 + y_3| \leq 3, \quad \sum_i y_i^2 + (\sum_i y_i)^2 \leq 48\}. \quad (4.1)$$

The section roots are $\pm t_1, \pm t_2, \pm t_3$ and $\pm(t_1 + t_2 + t_3)$. Inner products with these eight vectors give the linear bounds in (4.1), while $y^T H^{-1} y \leq 6$ gives its quadratic bound. Their own signatures form $E = \{\pm H e_1, \pm H e_2, \pm H e_3, \pm H(1, 1, 1)^T\}$. Each has multiplicity one: its section projection already has squared norm six. Enumeration of the displayed three-coordinate domain gives $|D_0| = 231$ and $|D_0 \cup E| = 239$.

For each pair $\{y, -y\} \subset D_0$, take the lexicographically smaller representative and set $v_{[y]} = f(y) = f(-y)$; retain zero separately. There are $(231 + 1)/2 = 116$ variables. For a multi-index a of even total degree at most ten, its moment row is

$$A_{a,[y]} = (2 - \mathbf{1}_{y=0}) \prod_i y_i^{a_i}, \quad b_a = M_a - \sum_{y \in E} \prod_i y_i^{a_i},$$

where M_a is the spherical moment defined above. There are $\sum_{j=0}^5 \binom{2j+2}{2} = 161$ such monomials. This constructs all entries of the 161×116 system $Av = b$.

The fibre count $f(y)$ counts shell vectors satisfying $(x \cdot t_1, x \cdot t_2, x \cdot t_3) = y$. The target is $f(e_1) + f(e_2) + f(e_3)$, not every individual multiplicity. After moving known exceptional contributions to the right, write the moment equations as $Av = b$, with $v \geq 0$. For a statistic

$c^\top v$, a rational vector w with $c - A^\top w \geq 0$ gives

$$c^\top v = w^\top b + (c - A^\top w)^\top v \geq w^\top b.$$

This is the form of the moment certificate. Choosing the statistic well makes an accessible fibre appear with positive coefficient in the lower bound, so a small explicit witness set becomes useful.

Proposition 4.2 (A positive fibre contribution). *The signature multiplicities satisfy*

$$f(e_1) + f(e_2) + f(e_3) \geq 7377408 + 9f(-2, -2, 3).$$

Proof. Give c coefficient $+1$ on the three classes $[e_i]$, coefficient -9 on $[(-2, -2, 3)]$, and zero elsewhere. Order the representatives and multi-indices lexicographically. The multiplier in `dual.json`, supplied in Appendix B, then has 161 rational entries and satisfies, by exact matrix multiplication, $c - A^\top w \geq 0$ and $w^\top b = 7377408$. The preceding identity with $v \geq 0$ gives the assertion. $\square$

Corollary 4.3 (The 45-dimensional construction). *The specified quadratic-residue neighbour and section give $K(45) \geq 7380090$.*

Proof. The certificate supplies 298 distinct integer lattice vectors of squared norm six and signature $(-2, -2, 3)$. Thus the right side is at least 7,380,090.

Project the three fibres e_1, e_2, e_3 onto the section's orthogonal complement. The projection formula is $\pi(x) = x - \text{proj}_U x$, giving

$$\pi(x) \cdot \pi(x') = x \cdot x' - y^\top H^{-1} y'.$$

For $y = e_i$, the squared section projection norm is $1/4$; for distinct e_i, e_j , their section projections have product $1/8$. Since distinct mother-shell vectors have product at most three, each projected squared norm is $23/4$. Products are at most $11/4$ within a fibre and $23/8$ between different fibres. Dividing by $\sqrt{23/4}$ gives a kissing configuration:

$$K(45) \geq 7377408 + 9(298) = 7380090.$$

$\square$

A witness search in a Pless lattice gave 292 vectors and the bound 7,380,036. Enlarging the choice of anchors led to 298 vectors in a quadratic-residue even neighbour, with each additional witness contributing nine points to the projected bound. To organize the search, fix a norm-eight anchor $p \in \Lambda$ and set

$$A_p = \{x \in X : x \cdot p = 4\}.$$

This parent set has exactly 2256 points, with completeness certified by low-degree moments. Since $x \pm p$ are nonzero lattice vectors, minimum six gives $|x \cdot p| \leq 4$. The polynomial

$$P(t) = t^2(t^2 - 1)(t^2 - 4)(t^2 - 9)$$

vanishes at all allowed integral products except ± 4 , where it equals 20160. The eighth spherical moment gives $\sum_{x \in X} P(x \cdot p) = 90961920$. Antipodal symmetry makes the two endpoint fibres equal, hence $|A_p| = 90961920/(2 \cdot 20160) = 2256$. The involution $x \mapsto p - x$ preserves A_p .

The graph vertices are its 1,128 pairs $\{x, p - x\}$ centred at $p/2$. Two pairs are adjacent when representatives have inner product one or three; exchanging an endpoint preserves this condition. Fibre searches become common-neighbour calculations in this graph.

Use the integer numerator coordinates $p = y/\sqrt{12}$ of the appendix. The selected y has six coordinates of absolute value three and 42 of absolute value one, so $y^2 = 96$. This odd-coordinate shape explores another part of the neighbour rather than repeating anchors from its integral sector. The resulting graph has degree 368 and a selected nonadjacent pair with 70 common neighbours. The difference $368 - 70 = 298$ identifies the witness fibre. The correspondence is explicit. Denote the selected endpoints by r, s and use the section $T = (-s, s - p, p - r)$. For a pair adjacent to the vertex of r but not that of s , choose its endpoint a with $r \cdot a = 3$; then $s \cdot a = 2$. Set $w = p - a$. It satisfies

$$\|w\|^2 = 6, \quad (-s \cdot w, (s - p) \cdot w, (p - r) \cdot w) = (-2, -2, 3).$$

Different pairs give different witnesses, so the 298 selected graph vertices yield precisely the required lattice vectors. The parent-graph checker in Appendix B reconstructs the section and exactly the 298 vectors in the final certificate. The rational moment inequality then gives the stated bound.

5 Construction principles and further questions

The point counts follow from specific changes in finite structure. In the code constructions, an exchange is governed by the union of conflicts; allowing signed local replacements enlarges the set of available changes. In the equatorial construction, moving to a different shell supplies directions that remain compatible with all existing caps. In the section constructions, the ambient embedding determines projection multiplicities that are invisible in the abstract Gram type alone.

These observations suggest concrete mathematical variables for further work. The support family and sign assignments can change together. Head classes and tail labels can be chosen by their net contribution to the whole configuration. An embedded section can be studied through a larger parent root system. A moment identity can be selected for the statistic that is easiest to witness, rather than for a full reconstruction of every fibre.

The accompanying finite data preserve these choices as explicit objects. They make it possible to revisit a construction, test a new local change, and determine its effect on the final count without repeating the entire discovery search.

A The minimum of the ambient lattice used in dimension 45

Let C be the extended ternary quadratic-residue $[48, 24, 15]$ code [21], and put $M = \{z \in \mathbb{Z}^{48} : z \bmod 3 \in C\}$ and $L_0 = \{z/\sqrt{3} : z \in M, z^2 \equiv 0 \pmod{6}\}$. The supplied basis determines a characteristic numerator $\chi \in M$ with odd coordinates, $\chi^2 = 264$, and $\chi \cdot z \equiv z^2 \pmod{6}$. Define

$$\Lambda = L_0 \cup (\chi/(2\sqrt{3}) + L_0).$$

The characteristic congruence and $\chi^2/12 = 22$ give integrality and evenness; the two index-two constructions give covolume one. The program reconstructs an integer numerator basis B and verifies $BB^T = 12G$ for the exact Gram matrix used by the 298-vector certificate.

In L_0 , squared norm two or four would require $z^2 \leq 12$. The code minimum weight 15 forces $z \bmod 3 = 0$, whose positive even lattice norms are at least six. In the other coset, $y = \chi + 2z$ has 48 odd coordinates, so $y^2/12 \geq 4$. Equality requires $y_i = \pm 1$.

Let $s_i \in \{\pm 1\}$ represent $\chi_i \bmod 3$. The finite checks give $s \in C$, $\chi \cdot s = 66$, and $\chi_i s_i \equiv 1 \pmod{6}$. Normalizing coordinates by s gives a code containing the all-one word. Its two-symbol enumerator is $1 + 94t^{24} + t^{48}$ [22, equation (20)], so binary-valued codewords have weights 0, 24 or 48. Writing $y = s(1 - 2b)$ gives such a binary word of weight w , but

$$\chi \cdot z = \frac{\chi \cdot s - \chi^2}{2} - \sum_i \chi_i s_i b_i \equiv -99 - w \equiv 3 \pmod{6},$$

contrary to $z/\sqrt{3} \in L_0$. Evenness excludes all other positive norms below six, while $\sqrt{3}(e_1 + e_2)$ has squared norm six. The theta-series and design-moment arguments therefore apply to this specified lattice.

B Construction data and exact verification

This PDF contains the attachment `certificates.zip`. Save it from the PDF viewer's attachment panel, or extract it with `pdftdetach -saveall main.pdf`. It contains the complete finite inputs and the programs used to check them. The English and Chinese editions embed identical archives.

Each construction combines a geometric argument covering all pair types with finite input satisfying its hypotheses. For the code constructions, the input consists of supports, generators, coset representatives and signed vectors. For the lattice constructions, it consists of integer directions, Gram matrices, embedded sections and rational moment certificates. The catalogue `constructions/catalog/results.json` specifies the defining files, exact count and verification tasks for each dimension. The following paths are relative to `constructions/`.

For dimensions 32–37 and 39, `codes/data/` contains the complete support lists, binary generators, coset representatives and exchange records. The local replacements for dimensions 35 and 36 are in `codes/hadamard/data/`; their generator file and the reconstruction program reproduce the stored signed vectors exactly. Dimension 37 has its own 512-point replacement and coordinate map in `codes/data/`.

For dimension 38, `d38/base-data/` specifies the Golay generator, Leech line classes, tail triangles and relative rotation defining the 591,612-point base. The 144 added representatives are in `d38/added-lines.txt`. The checker reconstructs the shell, verifies the base, tests the Leech congruences of every new vector, and checks all new mutual and boundary products in exact arithmetic.

For dimension 43, `sections/d43/data/parent-section.txt` gives the ambient Gram matrix and the eight rows embedding the numbered E_8 roots. `sections/d43/data/certificate.json` contains the orbit data, rational moment matrix and multiplier determining the D_5 count. One program reconstructs this system from the geometry; another checks the rational matrix identity independently.

For dimension 45, `sections/qr/` reconstructs the extremal neighbour. Within `sections/d45/data/`, `record.json` contains the section and 298 witnesses, `dual.json` the exact multiplier of Proposition 4.2, and `parent-vectors.json` the 2,256-point parent set. The checkers regenerate

the moment domain and equations, verify lattice membership and inner products, and recover the same witnesses from the parent graph.

After extracting the archive, its `README.md` gives the dependencies and commands. With SageMath, NumPy, SymPy and a C++ compiler installed, `sage -python tools/reproduce.py --suite all --output ../verification` runs every listed check in a separate directory, without network access. The output records each check and its inputs. The archive's manifest identifies every distributed file; the full repository also contains the report sources and the recorded verification results.

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